

HyperStar / GRRLE Synthetic Benchmark

A reproducible validation study for relational closure and graded model support

Abstract. This benchmark operationalizes the econometric HyperStar proposal as an explicit closure-aware support score and tests it on four synthetic tasks: OLS, instrumental variables, difference-in-differences, and an AR(1) state-space proxy. Eighty seeded replications per task compare closure residuals, held-out prediction error, and GRRLE support. The results validate a reproducible pipeline and expose an important distinction: a model can satisfy a target closure condition while another model predicts better. These are pipeline results, not evidence of general empirical superiority.

1. Protocol

For each candidate model, the benchmark records a task-specific closure residual and held-out mean squared error. Evidence is defined transparently as $\exp(-\text{normalized closure}) \times \exp(-\text{normalized held-out MSE})$. The evidence is then iterated through a normalized relaxation rule with persistence $\rho = 0.65$. The scoring rule is a research prototype, not a universal estimator.

2. Results

Task	Model	Closure mean	MSE mean	Support win	Prediction win
DiD	DiD	0.022	0.453	100%	100%
DiD	naive	0.412	4.097	0%	0%
IV	IV	0.000	6.475	100%	0%
IV	OLS	0.096	0.781	0%	100%
OLS	linear	0.000	1.476	0%	0%
OLS	quadratic	0.000	0.638	100%	100%
state-space	AR1	0.490	0.490	100%	100%
state-space	mean	1.241	1.258	0%	0%

Table 1. Means across 80 seeded replications per task. “Support win” is the share of replications in which the relaxation support ranking selected the model; “prediction win” is the share in which it had the lowest held-out MSE.

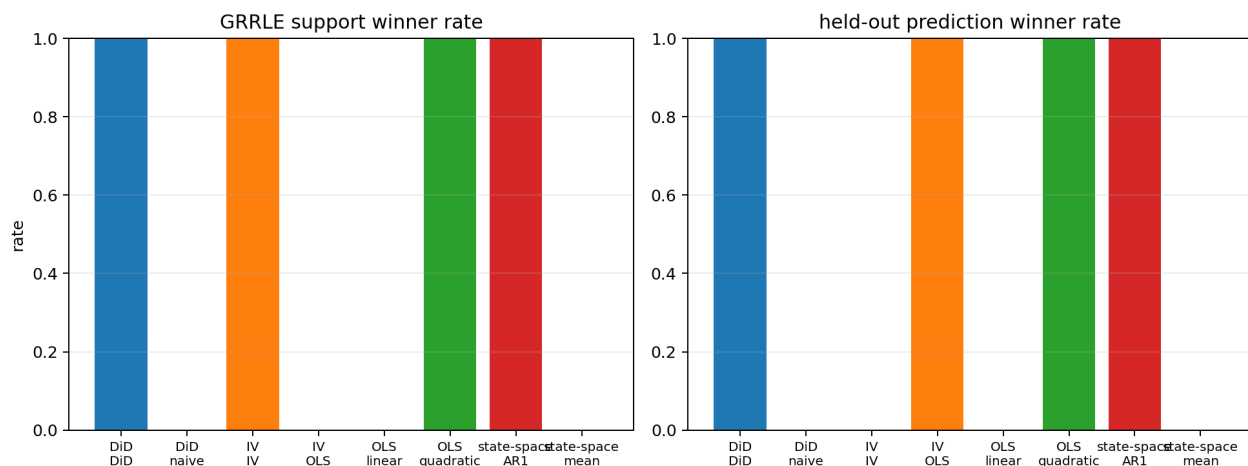


Figure 1. Support and prediction winners. The IV task separates causal-moment closure from predictive performance: IV satisfies the instrument moment by construction, while OLS predicts this endogenous synthetic outcome better.

3. Interpretation

Quadratic OLS, DiD, and AR(1) were selected by both closure-aware support and held-out prediction in this design. IV was selected by closure support but not by prediction. That disagreement is scientifically useful: closure is relative to an estimand and a declared restriction, whereas prediction is a different objective. A low residual cannot establish identification, causal validity, or transportability without additional assumptions.

4. Limits and next test

The data-generating processes are synthetic, the candidate sets are small, and the support score includes held-out error by design. The next study should preregister public-data benchmarks, add weak and invalid instruments, vary DiD parallel-trend violations, compare against cross-validation, information criteria, model averaging, and robust specification curves, and report calibration and rank stability under reparameterization.

Reproduction: `python3 hyperstar_benchmark.py`

Source: Michael Emanuel Glover · 19 September 2026